## Part 1

Version 1
Input: Are these reviews positive or negative?
Output: The reviews are mixed overall, with both positive and negative feedback across all five products.

I analyzed the 110 reviews in your CSV file and grouped them by product based on their sentiment.

# Overall sentiment

| Product | Overall impression |
|---|---|
| Breeze Wireless Earbuds | Mixed, leaning negative |
| Cozy Knit Throw Blanket | Mixed, leaning positive |
| Hearth Electric Kettle | Mixed |
| Lumen Smart Bulbs (4-Pack) | Mixed, leaning negative |
| Trailmark Hiking Backpack | Mixed, leaning positive |

# Key takeaways

- Trailmark Hiking Backpack: Customers generally like its durability, comfort, and value. However, some report broken zippers, poor stitching, and misleading waterproof claims.
- Cozy Knit Throw Blanket: Customers appreciate its softness, warmth, and appearance. Common complaints include shedding, shrinking, and quality issues.
- Hearth Electric Kettle: Customers like its design, fast boiling, and temperature settings. However, reliability and safety concerns (leaking and hot handles) are significant negatives.

- Breeze Wireless Earbuds: Positive feedback highlights sound quality, comfort, and battery life. Negative reviews frequently mention connectivity, charging, and microphone problems.
- Lumen Smart Bulbs: Customers enjoy the colors, scheduling, and smart-home features. However, app crashes, Wi-Fi disconnections, and setup difficulties hurt satisfaction.

# Conclusion

There isn't one universally positive or negative product in this dataset. The backpack and blanket appear to receive the most favorable feedback, while the earbuds and smart bulbs have more noticeable recurring complaints.

My recommendation: Pay particular attention to recurring complaints rather than isolated negative reviews. Repeated problems with reliability, connectivity, and product quality are the clearest opportunities for improvement.

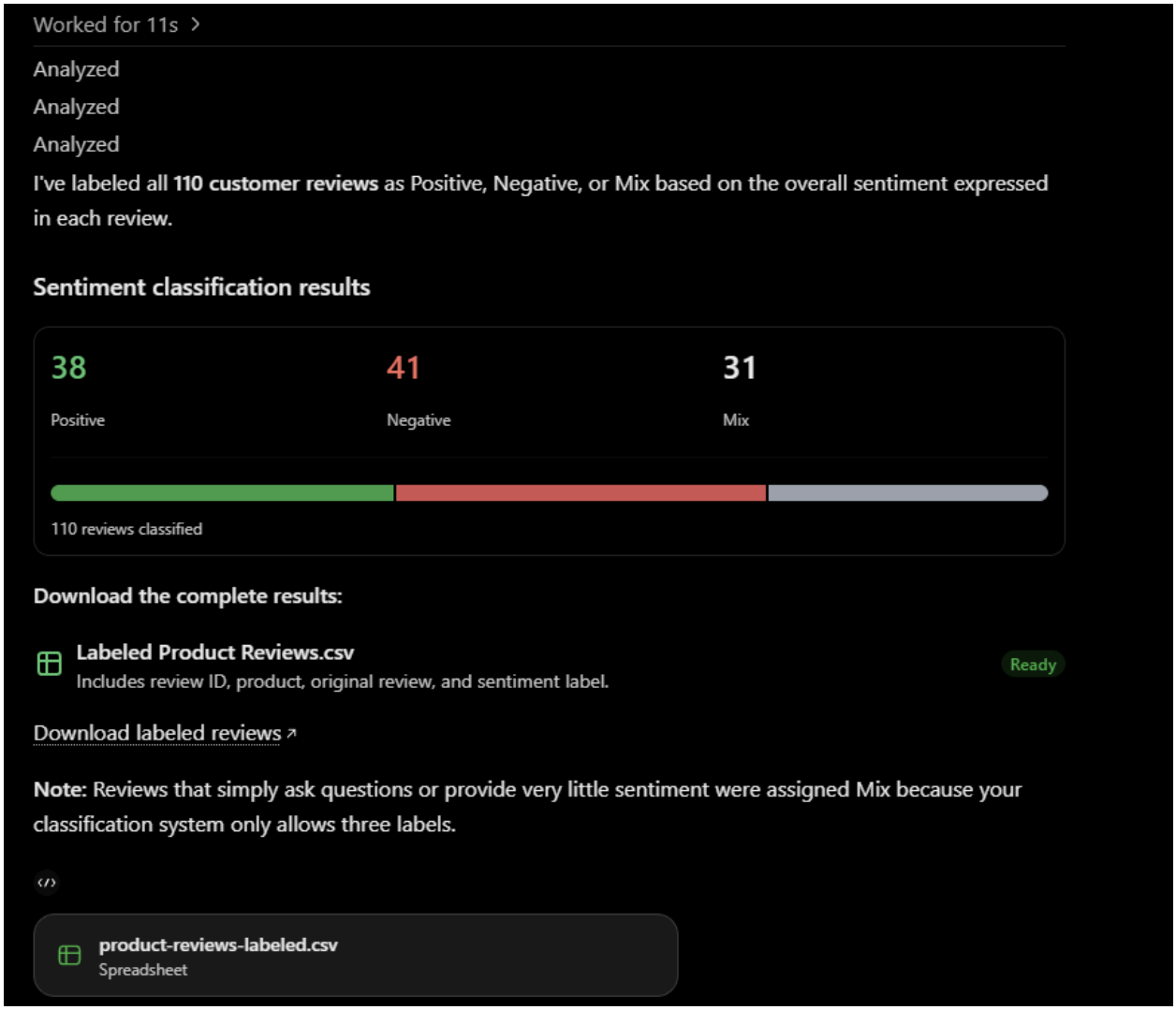
3. Read the output critically. Did the model label each review, or just summarize them? Could you sort or count the results? How did it handle reviews that aren't clearly positive or negative? The model provided a summary of the overall review of each product and provided some key takeaway instead of labeling each review. As the output is summary, there isn’t anything to sort or count and for review that aren’t clearly positive or negative, the model may have chosen wording such as leaning positive and leaning negative.

## Part 2

Version 2(Category + few-shot prompt)
Input: Please label each review as positive, negative, or mix. Positive means the customer is satisfied with the product with no major complaints. Negative means the customer is dissatisfied with the product with few good things to talk about the product. Mix means that there is an even split with the good and bad things the customer has to say about the product.

Here are a few examples of how to label each review:
Smells like chemicals out of the box. Took a week to air out. - Negative
Noise isolation is decent, but it's not true noise cancelling like the description implies. Still good for the gym. - Mix
Perfect for my commute. Clear calls, solid battery. - Positive

Output:




Version 2 output csv provided in the same folder.

Version 3(Role prompting + constraint)
Input: You are in charge of reviewing customer feedback to see if the customer likes the product or not. Are these reviews positive or negative for the product reviews csv file provided.

Rule:

- Label each review
- Provide reasoning in under 10 words for each review

Output:

Worked for 17s >

Analyzed

Analyzed

I reviewed all **110 customer reviews** and classified each as **Positive** or **Negative** based on the customer's overall experience with the product.

Each explanation is under 10 words. For reviews containing both praise and complaints, I used the dominant sentiment.

| Review ID | Sentiment | Reason |
|---|---|---|
| R001 | Negative | Strong chemical smell required a week to disappear. |
| R002 | Positive | Decent noise isolation and good gym performance. |
| R003 | Negative | Earbud stopped charging; customer support never responded. |
| R004 | Negative | Uncomfortable straps caused shoulder discomfort. |
| R005 | Negative | Not warm enough for winter use. |
| R006 | Positive | Clear calls and solid battery life. |
| R007 | Positive | Good value for the money. |
| R008 | Negative | Bulbs frequently disconnect from Wi-Fi. |
| R009 | Positive | Great bulbs despite a frustrating app. |
| R010 | Positive | Reliable performance over a year. |
| R011 | Negative | Blanket arrived with a stain. |
| R012 | Neutral | Asks about compatibility without expressing an opinion. |
| R013 | Positive | Replacement resolved the initial malfunction. |
| R014 | Positive | Customer service quickly replaced the damaged kettle. |
| R015 | Negative | Too small despite misleading product listing. |
| R016 | Neutral | Describes the product as merely fine. |

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|---|---|---|
| R104 | Neutral | Describes basic functionality without expressing satisfaction. |
| R105 | Positive | Perfect performance despite limited color selection. |
| R106 | Positive | Fast boiling and attractive appearance. |
| R107 | Negative | Poor quality caused unraveling within one month. |
| R108 | Negative | Frequent disconnections started after only three days. |
| R109 | Negative | Cheap charging case broke within one month. |
| R110 | Negative | Brightness falls short of advertised expectations. |

**Summary**

| 47 | 51 | 12 |
|---|---|---|
| Positive | Negative | Neutral |

**Overall sentiment distribution**



110 total reviews · Percentages based on all reviews

**Key takeaway:** Negative feedback slightly outweighs positive feedback. Product reliability, durability, and functionality are recurring customer concerns.

Neutral reviews are those that don't provide enough information to determine whether the customer likes the product.

Version 4(Output format)
Input: Are these reviews positive or negative? Return a table with these columns: review_id, product, review_text, sentiment, and reason

Output:



Here is the complete analysis and classification of all 110 reviews from `product-reviews.csv`, categorized by sentiment (Positive, Negative, Mixed, or Neutral) along with the reasoning behind each classification.

| review_id | product | review_text | sentiment | reason |
|---|---|---|---|---|
| R001 | Trailmark Hiking Backpack | Smells like chemicals out of the box. Took a week to air out. | Negative | Complains about a strong chemical odor requiring a week to air out. |
| R002 | Breeze Wireless Earbuds | Noise isolation is decent, but it's not true noise cancelling like the description implies. Still good for the gym. | Positive | Satisfied overall and finds them good for the gym despite overstated noise cancellation. |
| R003 | Breeze Wireless Earbuds | Left earbud stopped charging after two weeks. Contacted support and never heard back. | Negative | Earbud hardware failure combined with unresponsive customer support. |
| R004 | Trailmark Hiking Backpack | Straps dig into my shoulders after an hour. Returning. | Negative | Uncomfortable fit causing physical pain, resulting in a product return. |
| R005 | Cozy Knit Throw Blanket | Not warm enough for winter. Fine as a decorative throw. | Mixed | Fails as a warm winter blanket, but acceptable as a decorative item. |
| R006 | Breeze Wireless Earbuds | Perfect for my commute. Clear calls, solid battery. | Positive | High satisfaction with call clarity, battery life, and daily commuting performance. |
| R007 | Trailmark Hiking Backpack | decer[illegible]g for the mone[illegible]t for serious | Positive | Good value for the price, despite minor |

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| | | | | |
|---|---|---|---|---|
| R102 | Cozy Knit Throw Blanket | Best throw blanket I've owned. Heavy, warm, and soft. | Positive | Highest endorsement for feel, warmth, and weight. |
| R103 | Breeze Wireless Earbuds | meh. sound is tinny and they hurt after an hour | Negative | Poor sound quality and physical discomfort. |
| R104 | Hearth Electric Kettle | It boils water. It's a kettle. | Neutral | Simple functional statement without emotional sentiment. |
| R105 | Hearth Electric Kettle | Wish it came in more colors. Otherwise perfect. | Positive | High satisfaction with operation, minor desire for more colors. |
| R106 | Hearth Electric Kettle | good kettle!!! fast and pretty | Positive | Enthusiastic praise for speed and aesthetics. |
| R107 | Cozy Knit Throw Blanket | Unraveling after a month. Poor quality. | Negative | Premature wear and unraveling material. |
| R108 | Breeze Wireless Earbuds | Wow, they lasted a whole three days before disconnecting every five minutes. Truly impressive engineering. | Negative | Sarcastic complaint highlighting rapid connectivity breakdown. |
| R109 | Breeze Wireless Earbuds | Charging case feels cheap and the lid snapped off within a month. | Negative | Low quality materials and broken hardware. |
| R110 | Lumen Smart Bulbs (4-Pack) | Not as bright as a regular 60W bulb despite the claim. | Negative | Fails to match stated performance/brightness specifications. |

## Part 3

| Review # | Review | My Label | Model Label(Version 2) |
|---|---|---|---|
| R002 | Noise isolation is decent, but it's not true noise cancelling like the description implies. Still good for the gym | mix | mix |

| | | | |
|---|---|---|---|
| R010 | Second one I've bought, gave the first to my mom. Still going strong after a year. | Positive | Positive |
| R011 | Arrived with a stain on it | Negative | Negative |
| R014 | Mine arrived dented and customer service shipped a new one the next day. The kettle itself is fine | Mix | Mix |
| R017 | Absolutely bombproof. Dragged it through brush, rocks, and rivers with zero damage. | Positive | Positive |
| R024 | wow so smart, it forgot all my settings after an update. genius | Negative(sarcasm) | Negative |
| R036 | Ordered the gray and got navy. Kept it anyway since it's nice | Mix | Mix |
| R039 | Bought this for my office. Coworkers keep asking where I got it | Positive | Positive |
| R049 | The handle gets dangerously hot. Burned my hand the first day | Negative | Negative |
| R062 | Shipping took 3 weeks | Negative | Negative |

The model’s review for version 2 matches my label exactly.

Analysis

For my version 2 prompt, I used defining category and few shot prompting. For version 3, I role prompting and constraint. For version 4, I defined the output format.

The technique that works the best based on the comparison to reviews I labeled is in version 2 where I combined a few shot prompting and defining categories. I think it did the best job labeling the review because we left very little to ambiguity in our prompt by defining what each category means and by providing some examples. In comparison to version 3 where I used role prompting and constraint, there was still a lot left in terms of ambiguity in the reviews and thus, a lot of reviews I would have put as mix, it decided to put positive or negative. Version 4 has a similar problem with version 3 where since I didn't define how reviews should be categorized nor did I provide any examples, reviews where I would have put as mix, it put as positive or negative.

Based on my testing, the model was able to label every review from my prompt besides the very first one where it only provided a summary. By incorporating the different techniques, the model is able to understand that it should be labeling the reviews instead of providing a summary.